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China's Embodied Copper Flow from the Demand-Side and Production-Side Perspectives

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Abstract: Copper is a critical mineral resource and plays a crucial role in social and economic development. In China, the world's largest copper consumer, copper footprints and embodied copper transfers among sectors have not been studied sufficiently. Combining an environmentally extended input-output model and complex network method, this paper systematically analyzes China's copper consumption embodied in the final demand and the production process. The research shows that (1) from the perspective of the final demand, the *Construction* sector is the largest driver of copper consumption, contributing 3.27 Mt in 2020, followed by the *Manufacture of General Purpose Machinery* sector (1.31 Mt). (2) Structural path analysis (SPA) shows that mainly the *Construction* sector drives copper consumption from the *Production and Distribution of Electric Power and Heat Power* sector, followed by the *Manufacture of Non-metallic Mineral Products* sector, and so on. Conversely, the *Production and Distribution of Electric Power and Heat Power* sector is the main initial sector in the supply chain, driven by the *Construction* sector, the *Manufacture of Non-metallic Mineral Products* sector, the *Smelting and Processing of Metals* sector, and so on. (3) From the perspective of production, the *Transport, Storage, and Postal Services* sector is an important transfer station transforming resources from the upstream sectors to the downstream sectors along with the transfers of embodied copper. The *Production and Distribution of Electric Power and Heat Power* sector is an important supplier for the downstream sectors. The *Construction* sector is an important consumer for the upstream sectors. The sectors including the *Smelting and Processing of Metals* sector, the *Manufacture of Chemical Products* sector and the *Manufacture of Non-metallic Mineral Products* sector function well as transformers, suppliers and consumers in the process of embodied copper transfers. (4) From the perspective of production, the embodied copper flow system can be divided into four groups that are closely linked. E & C community is the core member of the whole embodied copper flow network. C and S community is the main consumer of embodied copper resources in the network, exporting a great deal of embodied resources from other communities. Finally, some policy proposals on the rational utilization of copper resources are put forward.

Keywords: embodied flow; copper; input-output; complex network; China



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1. Introduction

Copper is a strategic resource supporting economic and social development, which plays a vital part in many fields, such as power [1,2], equipment manufacturing [3–5], construction [6,7], new material industry [8,9], and so on. In the context of human efforts to achieve carbon neutrality [10,11], the extensive application of copper in the fields of new energy vehicles has become a new highlight of the copper consumption market [12,13]. China is among the leading consumers of copper (Cu) in the world [14,15], consuming 1.198×10^2 million tons (Mt) in 2020 and responsible for fifty percent of the world's consumption.

Current research on copper resources mainly focuses on the trade of copper [16,17], supply and demand [18–20], material flow [21–23], resource recycling [24,25], copper prices [26,27]. However, existing studies have hardly focused on copper footprints and embodied copper transfers in China. What is more, few scholars treat the direct and indirect copper flow system as a complex network, which leads to the research gap in the structure of the embodied copper flow network in China.

As a classic embodied resource, embodied energy was the first to be investigated using the environmentally extended input-output model [28,29], which is the amount of energy used from the cradle to the grave for a good or service [30,31]. Subsequently, with the occurrence of global issues such as ecological imbalance, environmental pollution, and resource shortage, the research objects of embodied sources gradually expanded from embodied energy to embodied emissions [32–34], embodied water [35,36], embodied material and material footprints [37,38], and so on.

The structural path analysis (SPA) was proposed by Defourny and Thorbecke [39] as a basis of input-output analysis. The SPA method is applied to trace how final demand drives substance emissions/resources consumption and to identify critical transmission sectors along the supply chain path. Previous studies have analyzed the SPA of embodied CO₂ emissions [40,41], embodied mercury emission [42–44], embodied energy consumption [45,46], and so on. Complex network theory, developed from graph theory and network theory, is a superior approach to exploring the structure of complex systems such as power networks, transportation networks, input-output networks, and so on, and then inquiring into the relationship between the network's structure and function [47,48]. Recently, the complex network theory has been used to trace the embodied resource flows among industries, including identifying and analyzing the key sectors and communities within the system.

Combining an environmentally extended input-output model, the SPA method and complex network theory, this paper is the first to explore China's embodied copper, for both direct and indirect flow from the demand-side and production-side perspectives. The structure of this study has mainly three aspects. First, demand-driven copper consumption from each sector is estimated based on the input-output table and sector-wise consumption datasets. Second, the embodied copper flow via key supply chains is traced and the corresponding contribution is expounded by using the SPA method. Third, complex network theory is used to identify the critical sectors and main communities of the embodied copper flow network. Finally, policy suggestions are put forward for the industries related to copper usage in China.

The rest of this paper is structured as follows: Section 2 introduces methods and data; Section 3 provides the results of copper embodied in the final demand and the production process and explores the relevant findings; Section 4 presents conclusions and policy implications; and Section 5 points out the limitations of this study and further research.

2. Methods and Data

2.1. Environmentally Extended Input-Output Model

The environmentally extended input-output model [32,49] is used to investigate China's copper consumption embodied in industrial sectors, combining the input-output table with environment-related external parameters, such as sector-wise consumption intensity.

First, we need to obtain the sector-wise copper consumption intensity Q , which is a row vector calculated using Equation (1)

$$Q = h(\hat{x})^{-1} \quad (1)$$

The x denotes each sector's total output in China. The h is a row vector, which is the satellite account of the direct copper consumption of each sector in China. The hat

($\hat{\cdot}$) indicates the vector is diagonalized. The copper flow of China embodied in the final demand of each sector is calculated by Equation (2)

$$C = Q * L * \hat{Y} \quad (2)$$

L refers to the Leontief inverse matrix $(I - A)^{-1}$, where A refers to the technical coefficient matrix, and I refers to the identity matrix. a_{ij} is the element of the matrix A , which represents the direct input from the industry i to produce the unitary total output of the industry j . l_{ij} is the element of the matrix $(I - A)^{-1}$, which represents the direct and indirect input from the industry i to produce the unitary total demand of the industry j . The column vector Y represents the total final demand of all the industries. Hence, the element c_{ij} of the row matrix is the amount of copper consumption of the industry i embodied in the final demand of the industry j .

Copper consumption embodied in industrial sectors can be shown by Equation (3)

$$F_{ij} = Q * (I - A)^{-1} * X_{ij} \quad (3)$$

where the notation X denotes the intermediate flow matrix of the input-output table in China for the year 2020, and X_{ij} refers to the monetary flow from the industry i to the industry j .

2.2. Structural Path Analysis

SPA is adopted to trace the embodied copper flow within the inter-industrial supply chains of China [50,51]. From the demand perspective, the Leontief inverse matrix is usually expanded using the Taylor series approximation, which is shown as Equation (4)

$$L = (I - A)^{-1} = I + A + A^2 + A^3 + \dots + A^n, \quad \sum_{n \rightarrow \infty} (A^n) = 0 \quad (4)$$

Combining Equation (2) with Equation (4), embodied copper flow in China can be expressed by the environmentally extended input-output model as Equation (5)

$$\begin{aligned} C &= Q * (I + A + A^2 + \dots + A^n) * \hat{Y} \\ &= QI\hat{Y} + QA\hat{Y} + QA^2\hat{Y} + \dots + QA^n\hat{Y} \end{aligned} \quad (5)$$

where $QA^n\hat{Y}$ represents the copper consumption occurring in the n th production layer, which is derived in the starting industry because of the final demand on the final industry in each supply chain path. Learning from previous experience [45,52], we just trace SPA up to the third order in this paper, which accounts for 82% of the total copper consumption for the year 2020.

2.3. Complex Network Theory

Complex network theory is used to identify the critical sectors and community structure in the embodied copper flow network. The complex network contains nodes, and links associated with each other, shown as Equation (6)

$$G = (N, E) \quad (6)$$

where N represents the nodes, and E represents the edges. For this study, the industrial sectors are treated as the node and embodied copper flow among sectors is taken as the edge. The flow volume of embodied copper is treated as the weight of the edge, then a direct-weighted embodied copper flow network is built.

2.3.1. Network Centrality

Network centrality is indicated by the weighted degree centrality including weighted out-degree and weighted in-degree based on the size of the flow among different industrial

sectors. The former is the total output of a sector to other sectors, while the latter is the total input of a sector from other sectors. The indicators are shown as Equations (7) and (8):

$$WI_i = \sum_{j=1} Q_{ji} \quad (7)$$

$$WO_i = \sum_{j=1} Q_{ij} \quad (8)$$

where Q_{ij} (Q_{ji}) represents the volume of embodied copper transforming from sector i (j) to sector j (i).

Furthermore, distinguished from the conventional network method rooted in the shortest path, we adopt some additional new indices to indicate network centrality, Strongest Path (SP) betweenness, downstream closeness and upstream closeness, which is created by Xu and Liang [47].

The SP betweenness of a node in the network represents the ability of an intermediary to transform resources to supply the whole economy, which is defined as Equation (9):

$$B_i = \sum_{s=1}^n \sum_{t=1, t \neq i}^n X_t q_{st} \quad (9)$$

The downstream closeness of a node in the network represents the important role as a supplier towards downstream industries, which is defined as Equation (10):

$$C_i^D = \frac{1}{n-1} \sum_{j=1}^n X_j q_{ij} \quad (10)$$

The upstream closeness of a node in the network represents the important role as a consumer towards upstream industries, which is defined as Equation (11):

$$C_i^U = \frac{1}{n-1} X_j \sum_{i=1}^n q_{ij} \quad (11)$$

where X_t , X_i and X_j represent the outputs of industry t , i and j .

2.3.2. Community Structure

As for community, we can think that it is a group of nodes closely associated with each other, but these nodes are sparsely connected in different communities [53,54]. In this paper, we adopt the detection algorithm to separate the network into communities, which was invented by Blondel et al. [55]. The modularity Q lies in the value between -1 and 1 , which can be shown as Equation (12):

$$Q = \frac{1}{2m} \sum_{i,j} \left[w_{ij} - \frac{A_i A_j}{2m} \right] \delta(c_i, c_j) \quad (12)$$

where w_{ij} refers to the weight of the edge between node i and node j .

2.4. Data Source

In this paper, two types of data are needed: China's non-competitive input-output table and China's sector-wise copper consumption data in 2020. The input-output table of 2020 is from the National Bureau of Statistics of China (<http://www.stats.gov.cn/>, accessed on 1 June 2022), which is the latest release in 2022. Names and corresponding abbreviations of all the sectors in China's input-output table are shown in Appendix A Table A1 at the end. Industrial names will be still used in the matter below for clarity. Instead of industrial names, the abbreviations will be used in the tables and figures below for brevity.

Copper refers to the refined copper in this paper, and the data about China's sector-wise copper consumption in 2020 act as a supplement to the industrial sectors (in physical units, Appendix A Table A1), which is from China's Copper Market Development Report in 2020 released by the Beijing Antaike Information Company [15].

3. Results and Discussion

3.1. Embodied Copper in the Final Demand

3.1.1. Embodied Copper in the Final Demand of Each Industry

In 2020, China's embodied copper in the total final demand was 10.935 Mt, which is the direct consumption in the production process for the whole economic system. In other words, embodied copper is driven by final demand. Table 1 illustrates the copper embodied in the final demand of each industry, calculated by using Equation (2). Embodied copper driven by the final demand of the *Construction* sector is the largest one, whose amount is 3.27 Mt and proportion is 29.92%, followed by the *Manufacture of General Purpose Machinery* sector (1.31 Mt, 11.94%), the *Manufacture of Transport Equipment* sector (1.16 Mt, 10.59%), the *Manufacture of Communication Equipment, Computers and Other Electronic Equipment* sector (1.05 Mt, 9.60%), and the *Production and Distribution of Electric Power and Heat Power* sector (0.76 Mt, 6.98%) and so on. These five sectors are also ones that directly consume copper (Appendix A Table A1), but the proportions of each industry are significantly different from the embodied copper driven by the demand. For example, the *Construction* sector directly consumes 0.98 Mt (proportion, 8.92%) and ranks fifth among these five sectors, while its final demand uses the most embodied copper.

Table 1. China's copper embodied in the final demand of each industry, proportion and embodied intensity in 2020.

Rank	Industry	Embodied Copper/Mt	Proportion	Embodied Intensity */g/CNY	Rank	Industry	Embodied Copper/Mt	Proportion	Embodied Intensity*/g/CNY
1	CON	32.72×10^{-1}	29.92%	0.121	22	AC	6.37×10^{-2}	0.58%	0.036
2	MGP	13.06×10^{-1}	11.94%	0.524	23	PTF	6.28×10^{-2}	0.57%	0.071
3	MTE	11.58×10^{-1}	10.59%	0.262	24	MPP	6.01×10^{-2}	0.55%	0.067
4	CCO	10.49×10^{-1}	9.60%	0.196	25	TI	5.78×10^{-2}	0.53%	0.066
5	EHP	76.30×10^{-2}	6.98%	1.221	26	WEP	5.49×10^{-2}	0.50%	0.070
6	MSP	28.50×10^{-2}	2.61%	0.124	27	FIN	5.33×10^{-2}	0.49%	0.015
7	EME	27.16×10^{-2}	2.48%	0.103	28	MNM	5.19×10^{-2}	0.47%	0.126
8	HS	24.20×10^{-2}	2.21%	0.050	29	SPM	4.95×10^{-2}	0.45%	0.121
9	PSS	23.46×10^{-2}	2.15%	0.031	30	PS	4.89×10^{-2}	0.45%	0.050
10	TSP	20.31×10^{-2}	1.86%	0.066	31	LCS	4.49×10^{-2}	0.41%	0.046
11	PFT	19.66×10^{-2}	1.80%	0.033	32	PDW	4.04×10^{-2}	0.37%	0.186
12	IT	19.52×10^{-2}	1.78%	0.042	33	CSE	3.24×10^{-2}	0.30%	0.034
13	MCP	16.91×10^{-2}	1.55%	0.090	34	PCN	2.32×10^{-2}	0.21%	0.061
14	MMP	13.70×10^{-2}	1.25%	0.137	35	MMI	2.15×10^{-2}	0.20%	0.092
15	SR	12.87×10^{-2}	1.18%	0.056	36	PDG	2.13×10^{-2}	0.19%	0.061
16	WRT	12.56×10^{-2}	1.15%	0.026	37	PME	0.85×10^{-2}	0.08%	0.143
17	AGR	11.62×10^{-2}	1.06%	0.027	38	WRO	0.48×10^{-2}	0.04%	0.036
18	LFF	11.32×10^{-2}	1.04%	0.055	39	MPN	0.41×10^{-2}	0.04%	0.123
19	EDU	10.85×10^{-2}	0.99%	0.021	40	MPM	0.16×10^{-2}	0.01%	0.116
20	RE	8.39×10^{-2}	0.77%	0.014	41	MWC	-0.10×10^{-2}	-0.01%	0.083
21	RRO	7.59×10^{-2}	0.69%	0.050	42	EPG	-0.41×10^{-2}	-0.04%	0.077

* Embodied intensity = the amount of embodied copper/the value added of one industry.

As for the embodied copper intensity (Table 1), the *Manufacture of General Purpose Machinery* sector ranks first among all the industries, with a value of 1.052 Gram/Chinese Yuan (g/CNY), followed by the *Manufacture of Transport Equipment* sector (0.652 g/CNY), the *Manufacture of Communication Equipment, Computers and Other Electronic Equipment* sector (0.631 g/CNY), the *Construction* sector (0.454 g/CNY), and the *Production and Distribution of Electric Power and Heat Power* sector (0.349 g/CNY), and so on.

3.1.2. Critical Path Analysis

To further reveal how the embodied copper flows from the initial producer to the final consumer in the economic system, SPA was used to identify the critical supply paths. Table 2 shows the top 30 supply chain paths with the greatest amount of embodied copper driven by the final demand in China (refer Supplementary Materials Table S2 for more results).

Table 2. Top 30 supply chain paths driving China's copper consumption in 2020.

Rank	Supply Chain Paths	Path Value	Order	Contribution Rate
1	CON	92.39	0	8.45%
2	MGP	86.99	0	7.96%
3	CCO	57.90	0	5.30%
4	MTE	56.82	0	5.20%
5	EHP	51.92	0	4.75%
6	EHP→CON	32.89	1	3.01%
7	EHP→MNM→CON	17.77	2	1.63%
8	MGP→MGP	16.82	1	1.54%
9	MTE→MTE	15.96	1	1.46%
10	CCO→CCO	15.57	1	1.42%
11	EHP→EHP	15.42	1	1.41%
12	EHP→SPM→CON	11.90	2	1.09%
13	EHP→EHP→CON	9.77	2	0.89%
14	MGP→CON	6.71	1	0.61%
15	MGP→MSP	6.41	1	0.59%
16	EHP→PSS	5.77	1	0.53%
17	EHP→MCP	5.40	1	0.49%
18	EHP→EHP→MNM→CON	5.28	3	0.48%
19	MGP→MTE	4.79	1	0.44%
20	EHP→MMP→CON	4.72	2	0.43%
21	EHP→EHP→EHP	4.58	2	0.42%
22	EHP→MCP→HS	4.54	2	0.42%
23	MTE→MTE→MTE	4.48	2	0.41%
24	CCO→CCO→CCO	4.19	2	0.38%
25	EHP→CCO	4.18	1	0.38%
26	EHP→MMP	3.78	1	0.35%
27	EHP→IT	3.76	1	0.34%
28	EHP→EHP→SPM→CON	3.53	3	0.32%
29	EHP→MGP	3.51	1	0.32%
30	EHP→TSP	3.51	1	0.32%

We found that the *Construction* sector mainly drives copper consumption from the *Production and Distribution of Electric Power and Heat Power* sector, followed by the *Manufacture of Non-metallic Mineral Products* sector, the *Smelting and Processing of Metals* sector, the *Manufacture of General Purpose Machinery* sector and so on. This is because the *Construction* sector requires electric power, cement (the *Manufacture of Non-metallic Mineral Products* sector), steel (the *Smelting and Processing of Metals* sector), and machinery equipment. What is more, the production of cement, steel, and equipment also needs electric power. It is consistent with the fact that all the above sectors need directly or indirectly using large amounts of copper in the production process.

As mentioned above (and in Appendix A Table A1), the *Production and Distribution of Electric Power and Heat Power* sector is the largest direct consumer. It can be seen that the *Production and Distribution of Electric Power and Heat Power* sector is the main initial sector in the supply chain, and it supplies electric power for several machinery equipment, the *Manufacture of Chemical Products* sector, the *Manufacture of Metal Products* sector, and the *Information Transfer, Software and Information Technology Services* sector, which shows these sectors are the main flow terminals of embodied copper.

3.2. Copper Embodied in the Production Process

For a better understanding of the embodied copper flow in the production process, the copper embodied among industrial sectors is calculated by Equation (3), and an embodied copper network is established as Equation (6). There are 42 nodes and 1661 edges in the embodied copper network (Figure 1), where the nodes refer to the 42 sectors, the edges refer to the copper embodied among the industrial sectors, the arrows refer to the direction of the embodied flow, and the thickness of the edges refers to the volume of the embodied copper flow.

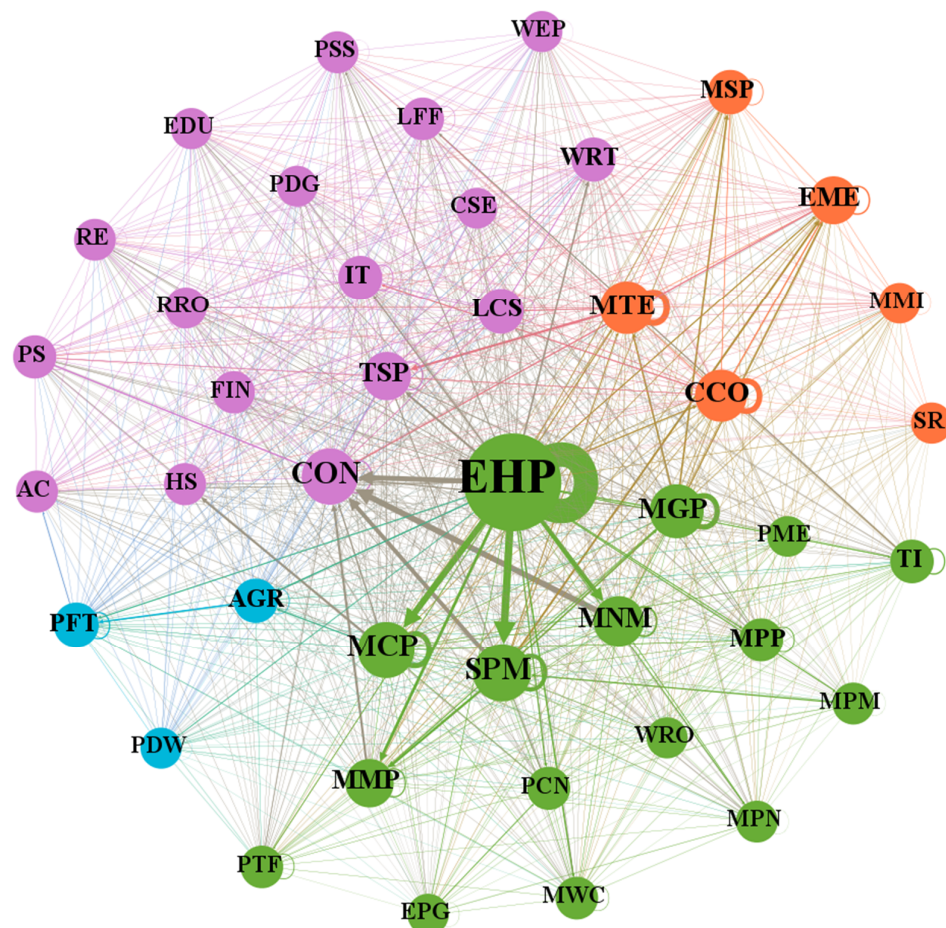


Figure 1. Embodied copper flow network of China in 2020.

3.2.1. Critical Sectors of the Embodied Copper Network Role Analysis Based on the Weighted Degree Centrality

The influence of a sector shows a positive correlation with its weighted degree [53,56–58]. Based on Equations (7) and (8), we can obtain the results as follows.

As shown in Figure 2 and Table S3 in Supplementary Materials, the top six sectors (the Production and Distribution of Electric Power and Heat Power sector; the Construction sector; the Smelting and Processing of Metals sector; the Manufacture of Chemical Products sector; the Manufacture of Transport Equipment sector; and the Manufacture of Communication Equipment, Computers and Other Electronic Equipment sector) import approximately 50% of the total imports of all sectors. As for the bottom six sectors (the Administration of Water, Environment, and Public Facilities sector; the Production and Distribution of Tap Water sector; the Culture, Sports, and Entertainment sector; the Other Manufacturing and Waste Resources sector; the Production and Distribution of Gas sector; and the Repair of Metal Products, Machinery and Equipment sector), the proportion of their import volume in all sectors' total imports is only 1.65%.

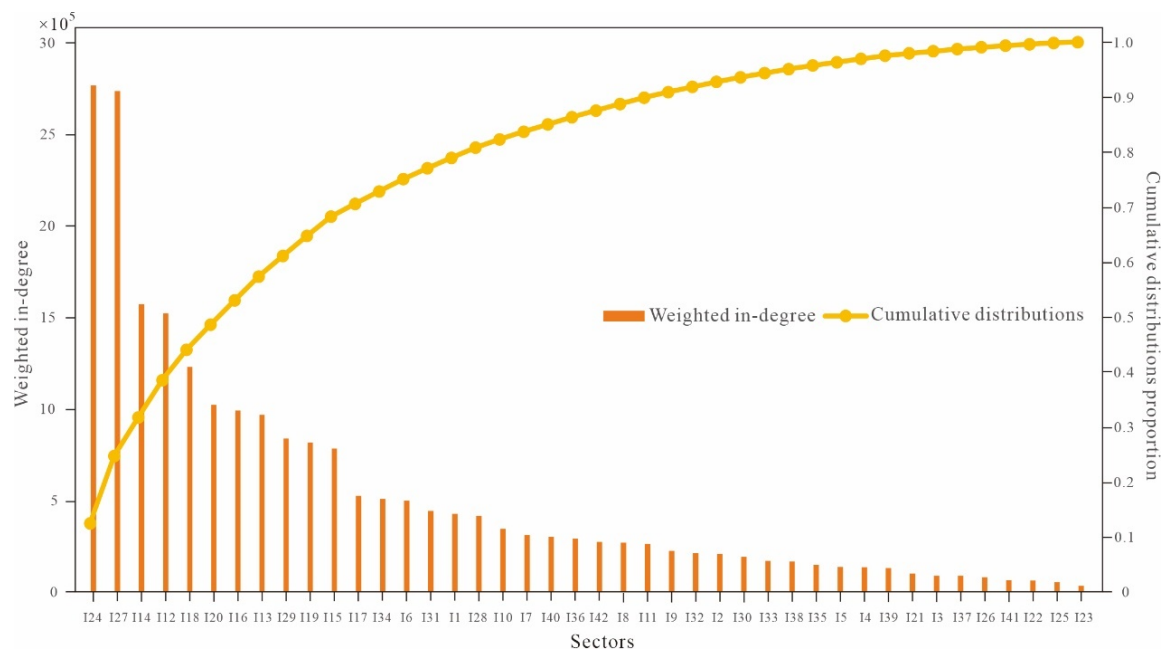


Figure 2. Weighted in-degree of all the sectors and their cumulative distributions.

From Figure 3 and Table S3 in Supplementary Materials, the top sector, the *Production and Distribution of Electric Power and Heat Power* sector, exports about 10% of the embodied copper. The top three sectors' exports, namely, the *Production and Distribution of Electric Power and Heat Power* sector; the *Smelting and Processing of Metals* sector; and the *Manufacture of Chemical Products* sector, account for nearly 50% of the total exports of all sectors. The bottom five sectors' sum exports, namely, the *Culture, Sports, and Entertainment* sector; the *Public Administration, Social Insurance, and Social Organizations* sector; the *Health Care and Social Work* sector; the *Education* sector; and the *Scientific Research* sector, take up only 0.22% of the whole exports of all sectors.

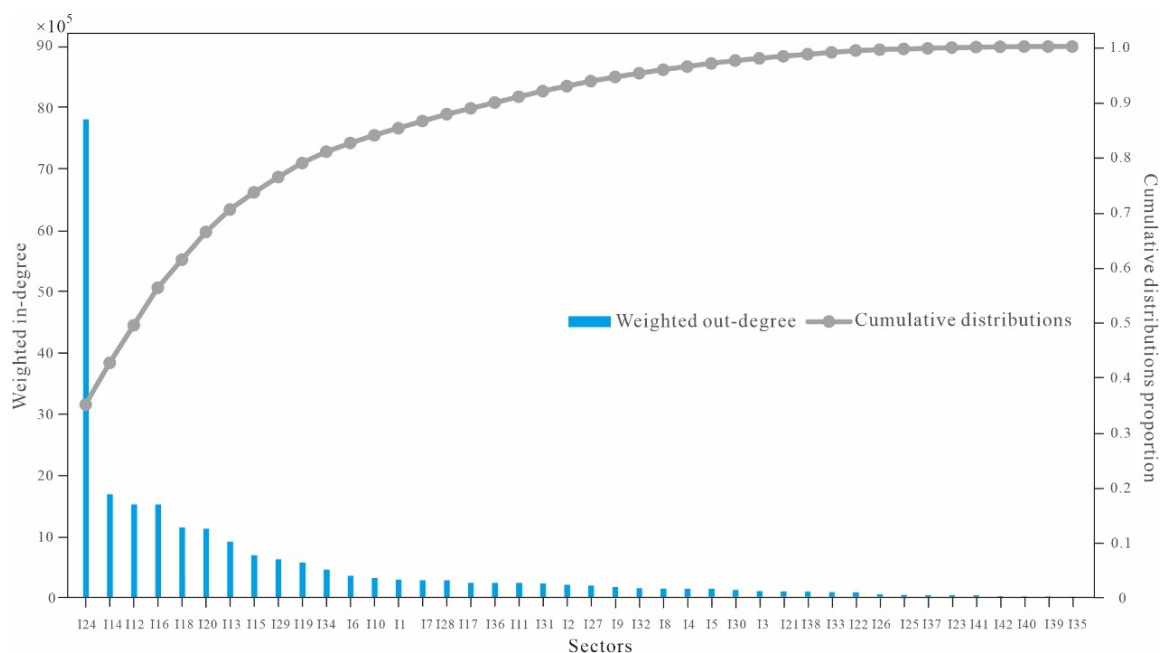


Figure 3. Weighted out-degree of all the sectors and their cumulative distributions.

According to the comprehensive comparison of the above results, we can find that, first, four sectors (the *Production and Distribution of Electric Power and Heat Power* sector; the

Smelting and Processing of Metals sector; the *Manufacture of Chemical Products* sector; and the *Manufacture of Transport Equipment* sector) are in the top six in both the imports and exports of the embodied copper flow, making up 32% and 55% for all sectors, respectively, which indicates their great impact on the import and export of embodied copper resources. Second, two sectors (the *Administration of Water, Environment, and Public Facilities* sector and the *Culture, Sports, and Entertainment* sector) are in the last seven with respect to the imports and exports of the embodied copper flow's, making up only 0.66% and 0.22% of all sectors, respectively, indicating their faint impact on the imports and exports of embodied copper flow. Third, there are five sectors whose sum import amount takes up half of the total import amount of all sectors' embodied copper flow, and three sectors whose sum export amount takes up 50% of total export amount of the embodied copper flow, which indicates the imports and exports of embodied copper flow are relatively concentrated in a few sectors.

Role Analysis Based on the SP Betweenness

As mentioned above, the SP betweenness of a node in the network represents the ability as a transfer station to transform resources. Based on Equation (9), we can obtain the results as follows (Table 3).

Table 3. The SP betweenness of China's each industry including the domestic rankings.

Sector	SP Betweenness	Rank	Sector	SP Betweenness	Rank
SPM	118,296	1	TI	6626	22
TSP	107,424	2	IT	4530	23
MCP	92,048	3	CCO	2885	24
MNM	61,293	4	MPN	1829	25
PCN	58,075	5	FIN	1502	26
LCS	53,300	6	WRO	1320	27
AGR	38,982	7	RE	932	28
EPG	30,678	8	PDG	728	29
PS	30,265	9	MMI	611	30
EHP	28,485	10	WEP	148	31
EME	25,160	11	PTF	97	32
MMP	21,750	12	MPM	0	33
MTE	18,191	13	MGP	0	34
LFF	16,993	14	PME	0	35
MPP	16,009	15	PDW	0	36
MSP	10,978	16	SR	0	37
WRT	10,857	17	RRO	0	38
CON	9962	18	EDU	0	39
AC	9907	19	HS	0	40
PFT	9362	20	CSE	0	41
MWC	7926	21	PSS	0	42

The *Smelting and Processing of Metals* sector, the *Transport, Storage, and Postal Services* sector, the *Manufacture of Chemical Products* sector, the *Manufacture of Non-metallic Mineral Products* sector, and the *Processing of Petroleum, Coking, Processing of Nuclear Fuel* sector are high on the list, while these sectors usually have no significant weighted degree. It indicates that these sectors play a connecting link role in the economic system and there are quite extensive supply and demand relations between them and other sectors.

Role Analysis Based on the Downstream Closeness

As mentioned above, the downstream closeness of a node in the network indicates its importance in the economic system as a supplier. Based on Equation (10), we can obtain the results as follows (Table 4).

Table 4. The downstream closeness of China's each industry including the domestic rankings.

Sector	Downstream Closeness	Rank	Sector	Downstream Closeness	Rank
EHP	128,533	1	IT	3240	22
SPM	26,880	2	EPG	3214	23
MGP	23,417	3	TI	3000	24
MCP	19,211	4	FIN	2959	25
MNM	17,273	5	AC	2200	26
MTE	13,472	6	PTF	1778	27
CCO	12,754	7	RRO	1771	28
MMP	12,080	8	RE	1732	29
TSP	11,722	9	MMI	1692	30
LCS	8614	10	LFF	1528	31
EME	8537	11	CON	1403	32
WRT	6239	12	WRO	1321	33
PCN	5590	13	PDW	770	34
AGR	5256	14	PDG	628	35
MPM	5036	15	WEP	532	36
MWC	4974	16	PME	521	37
PS	4258	17	CSE	432	38
MPP	4157	18	HS	120	39
MPN	3892	19	EDU	105	40
MSP	3729	20	PSS	87	41
PFT	3569	21	SR	0	42

The *Production and Distribution of Electric Power and Heat Power* sector ranks first by a large margin, indicating its great ability to supply resources for other sectors downstream along with the transfer of embodied copper. This result is consistent with the discussion about the SPA in Section 3.1.2 and the weighted degree analysis in the first part of Section 3.2.1.

The sectors, the *Smelting and Processing of Metals* sector, the *Manufacture of General Purpose Machinery* sector, the *Manufacture of Chemical Products* sector, and the *Manufacture of Non-metallic Mineral Products* sector, appear later in the succession. It indicates that these sectors have relatively strong ability as suppliers in the production system.

Role Analysis Based on the Upstream Closeness

As mentioned above, the upstream closeness of a node in the network indicates its importance in the economic system as a consumer. Based on Equation (11), we can obtain the following results (Table 5).

The *Construction* sector ranks first by a large margin, indicating its great ability to consume resources from other sectors in the upstream along with the transfers of embodied copper, which is also consistent with the result of the SPA in Section 3.1.2 and the weighted degree analysis in the first part of Section 3.2.1.

The sectors, the *Smelting and Processing of Metals* sector, the *Manufacture of Chemical Products* sector, the *Manufacture of Non-metallic Mineral Products* sector, and the *Transport, Storage, and Postal Services* sector, appear later in the succession, which indicates that these sectors have relatively strong ability as consumers in the production system.

3.2.2. Community Structure of the Embodied Copper Network

The community structure is the reflection of network heterogeneity [54,57], where communities generally consist of nodes tightly linked to each other [59,60]. According to the selection results of the community, targeted industrial regulations can be carried out. To identify communities, there are four main methods including the minimal tangential clustering algorithm [61], hierarchical clustering analysis [62], the maximum modularity algorithm [63], and Girvan–Newman Algorithm [64].

Table 5. The upstream closeness of China's each industry including the domestic rankings.

Sector	Upstream Closeness	Rank	Sector	Upstream Closeness	Rank
CON	60,866	1	EHP	5413	22
SPM	28,065	2	FIN	4870	23
MCP	23,784	3	LFF	4613	24
MNM	18,904	4	TI	4319	25
TSP	17,865	5	MWC	4185	26
EME	15,890	6	RE	4075	27
MMP	15,878	7	AC	3927	28
MTE	13,213	8	RRO	3792	29
MSP	10,857	9	PTF	3424	30
LCS	10,488	10	MPM	3215	31
WRT	10,481	11	SR	3207	32
MGP	10,100	12	MPN	3165	33
CCO	9800	13	EDU	2917	34
PFT	9051	14	EPG	2137	35
IT	8563	15	MMI	2078	36
AGR	8008	16	WEP	1934	37
HS	7472	17	PDW	1829	38
PCN	6435	18	CSE	1332	39
PSS	6226	19	PDG	1205	40
MPP	5809	20	WRO	1202	41
PS	5769	21	PME	746	42

In this paper, the maximum modularity algorithm is used to identify the community of embodied copper flow network. As can be seen from Table 6, the nodes and edges are marked with four colors, namely, the embodied copper flow network can be divided into four groups. Each community is named according to the type of sectors within. Blue represents the A and F community, which are dominated by the sectors, the *Agriculture, Forestry, Animal Husbandry and Fishery* sector and the *Food and Tobacco Processing* sector. Green represents the E and C community, which is dominated by the sectors, the *Production and Distribution of Electric Power and Heat Power* sector and the *Manufacture of Chemical Products* sector. Orange represents the E and R community, which is dominated by the sectors, the *Manufacture of Transport Equipment* sector and the *Manufacture of Electrical Machinery and Equipment* sector. Purple represents the C and S community, which is dominated by the *Construction* sector and the service industry.

Table 6. Relevant information about these four industrial communities.

	A & F	E & C	E & R	C & S
Sectors number (percentage)	3 (7.1%)	16 (38.1%)	6 (14.3%)	17 (40.5%)
Sum of embodied copper flow (percentage)	1.68×10^6 (3.8%)	26.2×10^6 (58.7%)	6.98×10^6 (15.6%)	9.79×10^6 (21.9%)
Sum of SP betweenness	4.8×10^4 (6.1%)	4.4×10^5 (55.8%)	5.8×10^4 (7.3%)	2.5×10^5 (30.9%)
Sum of downstream closeness	9.6×10^3 (2.7%)	2.6×10^5 (72.8%)	4.0×10^4 (11.2%)	4.8×10^4 (13.3%)
Sum of upstream closeness	1.9×10^4 (5.1%)	1.4×10^5 (37.3%)	5.5×10^4 (15.0%)	1.6×10^5 (42.6%)

To explore the role of the four industrial communities of the embodied copper flow network, we further analyze the number of sectors in each community, the sum flow of embodied copper, the sum of SP betweenness, the sum of downstream closeness, the sum of upstream closeness, and their respective percentages (Table 6). There are 3, 16, 6, and 17 sectors in the A and F community, the E and C community, the E and R community, and

the C and S community, respectively, which account for 7.1%, 38.1%, 14.3%, and 40.5% of the total number, respectively.

Figure 4 shows the relationships between the four communities from the perspective of the sum of the weighted degree, where the nodes indicate the communities. The connecting lines among the nodes represent the embodied copper flow and the arrows represent the flow's direction. The volume of the flow is in proportion to the thickness of lines. We can find that, as for the weighted degree, the SP betweenness and the downstream closeness, the E and C community occupies the absolute advantage, though the number of the sectors within ranks second to that of the C and S community. As for the sum of upstream closeness, the C and S community ranks first in the four communities, with a value slightly higher than that of the E and C community. As for the A and F community and the E and R community, the proportions of their embodied copper flow show a roughly positive correlation with the number of the sectors within each community.

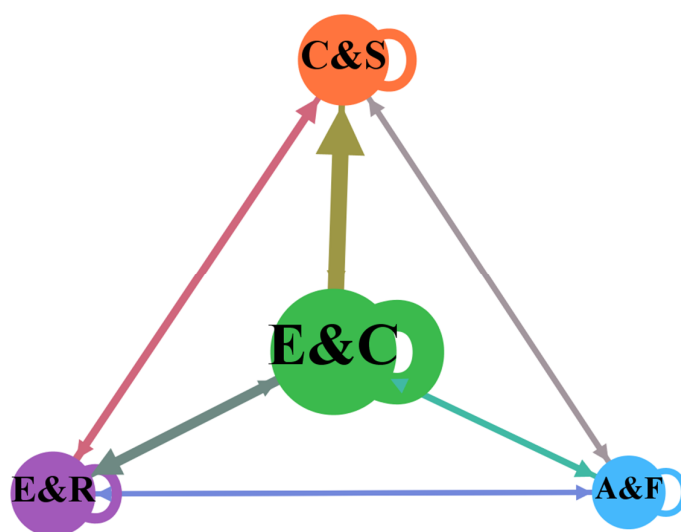


Figure 4. Embodied copper flow among different industrial communities.

To summarize, the results are found as follows. First, although the number of its sectors is not greater than that of the C and S community, the E and C community is the core member of the entire embodied copper flow network, which provides a large number of embodied copper resources to other communities. Meanwhile, it is also an important transmission center, receiving large amounts of embodied copper resources from other groups and exporting the same amount of resources to other communities simultaneously. Second, although the C and S community contains the most sectors, it is the main consumer of embodied copper resources, because its export volume of embodied copper resources is far less than its import volume. Third, each community has quite a strong self-circulating flow, i.e., self-consumption within each community is important.

4. Conclusions and Policy Implications

Some interesting conclusions are as follows:

1. From the perspective of final demand, different from the direct consumption, the *Construction* sector is the largest driver of copper consumption, contributing 3.27 Mt in 2020, followed by the *Manufacture of General Purpose Machinery* sector (1.31 Mt), and the *Manufacture of Transport Equipment* sector (1.16 Mt), the *Manufacture of Communication Equipment, Computers and Other Electronic Equipment* sector (1.05 Mt), and the *Production and Distribution of Electric Power and Heat Power* sector (0.76 Mt).
2. From the perspective of final demand, the SPA method shows that the *Construction* sector mainly drives copper consumption from the *Production and Distribution of Electric Power and Heat Power* sector, followed by the *Manufacture of Non-metallic Mineral Products* sector, the *Smelting and Processing of Metals* sector, the *Manufacture*

of *General Purpose Machinery* sector, and so on. On the other hand, the *Production and Distribution of Electric Power and Heat Power* sector is the main initial sector in the supply chain, driven by the *Construction* sector, the *Manufacture of Non-metallic Mineral Products* sector, the *Smelting and Processing of Metals* sector and so on.

3. From the perspective of production, the sectors including the *Transport, Storage, and Postal Services* sector and the *Processing of Petroleum, Coking, Processing of Nuclear Fuel* sector are important transfer stations, transforming resources from the upstream sectors to the downstream sectors along with transfers of embodied copper. The sectors including the *Production and Distribution of Electric Power and Heat Power* sector and the *Manufacture of General Purpose Machinery* sector are important suppliers for the downstream sectors. The sectors including the *Construction* sector and the *Manufacture of Electrical Machinery and Equipment* sector are important consumers for the upstream sectors. It is also worth noting that the sectors including the *Smelting and Processing of Metals* sector, the *Manufacture of Chemical Products* sector, and the *Manufacture of Non-metallic Mineral Products* sector perform very well in all the five lists including weighted degree, SP betweenness, downstream closeness and upstream closeness, indicating that they function well as transformers, suppliers and consumers in the process of embodied copper transfers.
4. From the perspective of production, there are four industrial communities derived from the embodied copper flow network. The E and C community is the core member of the whole embodied copper flow system, both as a critical supplier and as an important transmission center of the embodied copper resources. The C and S community is the main consumer of embodied copper resources in the network, exporting a great deal of embodied copper from other communities. What is more, there is quite a strong self-circulating flow in each community.

The following policy implications are proposed based on this study:

First, copper is a non-renewable resource and generally running out with from massive use. Against the background of carbon neutrality, copper is playing an increasingly important role in many fields such as renewable energy and electrification. Recently, there are more and more countries defining copper as a critical mineral. It has become quite essential to use copper resources reasonably and efficiently, especially in the global transition toward a renewables-powered society. Therefore, a high value should be attached to the critical sectors from the perspectives of the final demand and production. For the transforming center, such as the sectors including the *Transport, Storage, and Postal Services* sector and the *Processing of Petroleum, Coking, Processing of Nuclear Fuel* sector, we should enhance productivity to speed up the transmission of embodied copper flow to promote economic development. For the main initial sector in the supply chain or the main supplier of embodied copper, such as the sectors including the *Production and Distribution of Electric Power and Heat Power* sector and the *Manufacture of General Purpose Machinery* sector, we should improve the utilization efficiency of copper in the production process with technical optimization so as to conserve copper. For the main driver or consumer (receiver) of embodied copper, such as the sectors including the *Construction* sector and the *Manufacture of Electrical Machinery and Equipment* sector, we should adjust related tax policies to guide behaviors among consumers so as to avoid wasting copper.

As for the sectors taking on multiple roles in the economic system, including the *Smelting and Processing of Metals* sector, the *Manufacture of Chemical Products* sector and the *Manufacture of Non-metallic Mineral Products* sector, special attention is necessary. We should deal with them utilizing various means.

Second, these sectors in China's embodied copper flow system are closely linked, "one touch affects the whole body". Meanwhile, these are quite different supply chain paths and distinct groups containing several sectors with close relationships lying in the network. When we want to adjust a given sector, critical paths or industrial communities for overall planning should be taken into consideration.

For instance, for the path “*Production and Distribution of Electric Power and Heat Power* → *Manufacture of Non-metallic Mineral Products* → *Construction*”, attention should be paid to improving industrial coordination. At the same time, an accurate assessment of the investment in the downstream sectors such as the *Construction* sector should be implemented to reduce invalid demand for copper in the key supply chain path. When the *Construction* sector in the C and S community needs to be adjusted, we should consider its similarity with the *Wholesale and Retail Trades* sector, the *Transport, Storage, and Postal Services* sector and so on in the same community. If necessary, we can substitute one sector for another in the same community, or adjust these backbone sectors together.

5. Limitations of This Study and Further Research

First, there is a temporal hysteresis of China’s input-output table in 2020 to discuss the current embodied copper. Second, due to data availability and technical limits, China’s sector-wise copper consumption data are not perfect. We chose the data forming large groups and ignored the small ones because it was difficult to match them with the sectors.

In further research, we will adopt a more recent input-output table and more accurate sector-wise consumption data for copper to explore the embodied copper flow in China. We will also use the inter-country input-output table to probe the embodied copper flow around the world.

It should be noted that we explored China’s embodied copper flow in 2012, 2015, 2017, 2018 and 2020 to excavate the temporal evolution characteristics, but there are no changes in each index except for the increase in volume (refer to Supplementary Materials for more results).

The result of copper embodied in the final demand is in Table S1. The result of the critical path analysis is in Table S2. The result of critical sectors of the embodied copper network is in Table S3. The result of the community structure is in Table S4.

Supplementary Materials: The following supporting information can be downloaded at: <https://www.mdpi.com/article/10.3390/su15032199/s1>, Table S1: The result of copper embodied in the final demand; Table S2: The result of the critical path analysis; Table S3: The result of critical sectors of the embodied copper network; Table S4: The result of the community structure.

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Data Availability Statement: Data is provided in Supplementary Materials Tables S1–S4.

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Appendix A

Table A1. Names and corresponding abbreviations of sectors and direct consumption of copper.

Number	Industry	Abbreviation	Direct Consumption /Mt	Number	Industry	Abbreviation	Direct Consumption /Mt
1	Agriculture, Forestry, Animal Husbandry and Fishery	AGR	0	22	Other manufacturing and waste resources	WRO	0
2	Mining and washing of coal	MWC	0	23	Repair of metal products, machinery and equipment	PME	0

Table A1. Cont.

Number	Industry	Abbreviation	Direct Consumption /Mt	Number	Industry	Abbreviation	Direct Consumption /Mt
3	Extraction of petroleum and natural gas	EPG	0	24	Production and distribution of electric power and heat power	EHP	5.81
4	Mining and processing of metal ores	MPM	0	25	Production and distribution of gas	PDG	0
5	Mining and processing of nonmetal and other ores	MPN	0	26	Production and distribution of tap water	PDW	0
6	Food and tobacco processing	PFT	0	27	Construction	CON	0.975
7	Textile industry	TI	0	28	Wholesale and retail trades	WRT	0
8	Manufacture of leather, fur, feather and related products	LFF	0	29	Transport, storage, and postal services	TSP	0
9	Processing of timber and furniture	PTF	0	30	Accommodation and catering	AC	0
10	Manufacture of paper, printing and articles for culture, education and sport activity	MPP	0	31	Information transfer, software and information technology services	IT	0
11	Processing of petroleum, coking, processing of nuclear fuel	PCN	0	32	Finance	FIN	0
12	Manufacture of chemical products	MCP	0	33	Real estate	RE	0
13	Manuf. of non-metallic mineral products	MNM	0	34	Leasing and commercial services	LCS	0
14	Smelting and processing of metals	SPM	0	35	Scientific research	SR	0
15	Manufacture of metal products	MMP	0	36	Polytechnic services	PS	
16	Manufacture of general purpose machinery	MGP	1.86	37	Administration of water, environment, and public facilities	WEP	0
17	Manufacture of special purpose machinery	MSP	0	38	Resident, repair and other services	RRO	0
18	Manufacture of transport equipment	MTE	1.11	39	Education	EDU	0
19	Manufacture of electrical machinery and equipment	EME	0	40	Health care and social work	HS	0
20	Manufacture of communication equipment, computers and other electronic equipment	CCO	1.18	41	Culture, sports, and entertainment	CSE	0
21	Manufacture of measuring instruments	MMI	0	42	Public administration, social insurance, and social organizations	PSS	0

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